

By Newton's 3rd Law, force exerted by car has same magnitude as the normal contact force

10 Ans: B

$$g = \frac{GM}{R^2}$$

$$\begin{aligned}\text{Work done} = \text{increase in GPE} &= -\frac{GMm}{3R} - \frac{GMm}{R} \\ &= \frac{2}{3} \frac{GMm}{R} \\ &= \frac{2}{3} \frac{GMm}{R^2} R \\ &= \frac{2}{3} mgR\end{aligned}$$

11 Ans: D